

Semantic Prompt and Graph-Convolution-Structure Distillation Framework for Semantic Segmentation of Remote Sensing Images

Wujie Zhou^{ID}, Senior Member, IEEE, Jin Xie, and Caie Xu^{ID}

Abstract—High-resolution remote sensing semantic segmentation plays a critical role in land-use monitoring, urban planning, and disaster response. However, its deployment remains challenging owing to modality heterogeneity, fine-scale object structures, and the high computational cost of current deep learning models. To address these challenges, we propose a semantic prompt and graph-convolution-structure distillation framework (SPGSNet-S*), a compact, yet effective architecture that integrates multimodal feature enhancement with dual-path knowledge distillation (KD). Specifically, we design two lightweight modules—auxiliary spatial feature extraction (ASFE) and red–green–blue (RGB) representation—to denoise and align noisy normalized digital surface model (nDSM) features with RGB imagery, enabling robust feature fusion. In addition, we introduce a dual distillation scheme comprising graph-convolution-based structure distillation, which captures and transfers spatial topological dependencies, and semantic prompt distillation (SPD), which dynamically generates and injects class-aware visual prompts without external text supervision. Experimental results on the Vaihingen and Potsdam datasets show that SPGSNet-S* outperforms several state-of-the-art methods, achieving competitive performance with only 8.89 M parameters and 2.29 G floating-point operations (FLOPs). The source code and experimental results are publicly available at <https://github.com/110-011/SPGSNet>

Index Terms—Graph convolution (GCN), knowledge distillation (KD), prompt learning (PL), remote sensing images, semantic segmentation.

I. INTRODUCTION

SEMANTIC segmentation of high-resolution remote sensing images aims to assign a category label to each pixel, enabling a comprehensive and fine-grained understanding of image content. This technique plays a vital role in applications such as urban planning [1], disaster assessment [2], and environmental monitoring [3]. However, the task remains

challenging owing to complex object boundaries, significant scale variations, and severe class imbalance in remote sensing imagery. To incorporate geometric information, researchers often introduce normalized digital surface models (nDSMs) as an auxiliary modality. However, this introduces challenges such as modality heterogeneity and sensor misalignment, which complicate multimodal feature fusion.

Current studies on semantic segmentation in high-resolution remote sensing images generally follow two main approaches. The first approach focuses on single-modality segmentation methods based on convolutional neural networks (CNNs) [4], [5], [6], [7], [8] or Transformers [9], [10], [11], [12], [13]. These methods are effective at modeling local and global features; however, they often overlook geometric priors and struggle to handle ambiguous object boundaries. The second approach involves multimodal fusion methods, which typically integrate red–green–blue (RGB) imagery with auxiliary modalities, such as nDSMs or depth maps, using a dual-branch architecture or cross-modal alignment strategies [14], [15], [16], [17]. Although these methods generally achieve higher segmentation accuracy than single-modality methods, they often rely on strict modality alignment and overlook inherent cross-modal differences, limiting their robustness to the heterogeneity and noise present in real-world sensor data. In addition, many high-performance models have high computational costs and large parameter sizes, making them difficult to deploy on resource-constrained devices.

To balance accuracy and efficiency, knowledge distillation (KD) [18] has emerged as a widely adopted strategy in recent years. KD enables knowledge transfer from a powerful teacher model to a compact student model, allowing the student to inherit rich feature representations. However, while conventional KD methods are effective for model compression, they often rely on simple feature mimicry or soft-label supervision, overlooking structural dependencies and semantic abstraction in the teacher network.

Simultaneously, prompt learning (PL) [19] has gained attention as a promising approach for incorporating prior knowledge through compact guiding signals. PL has shown potential in steering pretrained models, particularly in natural language-guided tasks. However, most existing methods either depend on textual prompts and pretrained models such as CLIP [20] or are primarily designed for classification tasks [21]. Visual prompt-based distillation strategies for dense prediction in remote sensing remain limited, especially in scenarios without textual labels and where strong spatial–structural

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understanding is required. This raises a critical question: Is it possible to construct a distillation framework that avoids textual input while effectively integrating both structural and semantic prompts?

To address key challenges in remote sensing imagery—such as modality misalignment, difficulties with structural abstraction, and deployment constraints—we propose a unified semantic segmentation framework called SPGSNet. This framework combines multimodal fusion with a dual-channel KD strategy. It includes a high-capacity teacher model (SPGSNet-T) and a lightweight student model (SPGSNet-S*), both of which adopt a parallel RGB–nDSM dual-branch architecture to fully leverage complementary spectral and geometric information.

Our SPGSNet framework represents a systematic innovation that synergistically integrates multimodal feature enhancement, spatial–structural preservation, and semantic knowledge transfer. Unlike existing approaches that tackle these challenges in isolation, our unified framework establishes a coherent KD pipeline: 1) the auxiliary spatial feature extraction (ASFE) module and the representation module create a robust foundation for multimodal alignment; 2) graph-convolution-based structure distillation (GCSD) preserves spatial–structural dependencies during encoding-stage distillation; and 3) semantic prompt distillation (SPD) transfers semantic abstractions during decoding-stage distillation. This end-to-end design ensures that structural and semantic knowledge are complementarily transferred while maintaining spatial fidelity—a critical requirement in remote sensing applications where fine-grained boundaries and semantic consistency are essential.

To address noise and heterogeneity in nDSM data, we design an ASFE module. This module integrates bilateral filtering, spatial attention, and a multiscale channel recalibration mechanism to denoise, align, and reweigh nDSM features. The ASFE module outputs calibrated RGB and nDSM features and generates a spatial guidance map that captures geometric priors; these outputs are then passed to the fusion modules. In addition, we propose an RGB refinement module (RRM), a context-enhanced fusion unit that refines RGB representations under ASFE guidance. The RRM incorporates graph convolutional (GCN) attention, multiscale local feature separation, and an inverse attention mechanism to enhance spatial perception. This design enables the student model to delineate fine-grained boundaries and small object regions more accurately, improving robustness in complex remote sensing scenarios.

For KD, we construct a dual-path knowledge transfer framework. Furthermore, we introduce a graph-based convolutional structure distillation module in the encoding stage; it uses region-level graph modeling to capture and transfer spatial structural dependencies from the teacher model. Unlike previous methods that rely on global average pooling or feature vectorization, GCSD performs structural reasoning directly in the 2-D spatial domain, preserving the spatial topological information critical for pixel-wise segmentation.

In the decoding stage, we design the SPD module as a high-level semantic guidance mechanism. SPD dynamically generates semantic prompt maps from the decoding features of both teacher and student models and injects them into the

student decoder to guide the extraction of abstract category semantics. This mechanism operates independently of textual prompts or large-scale pretrained models, making it particularly suitable for remote sensing tasks where paired textual annotations are typically unavailable.

Through the synergistic optimization of these modules, the student model SPGSNet-S* effectively narrows the performance gap with respect to the teacher model, achieving high segmentation accuracy while significantly reducing the number of model parameters and computational overhead. The overall framework of the proposed SPGSNet with KD is presented in Fig. 1.

From 2024 to 2025, several multimodal fusion and structure-aware segmentation networks have been proposed for remote sensing imagery, focusing on cross-modal attention, dynamic fusion, and lightweight design (e.g., CATNet [13], STRD-Net [16], and CAInet [17]). While these methods improve fusion effectiveness or efficiency, they generally treat multimodal fusion and KD as independent problems. By contrast, our work integrates noise-robust multimodal fusion and structure–semantic-aware distillation into a unified framework, addressing both representation learning and deployment efficiency.

The main contributions of this article are as follows.

- 1) We propose a dual-branch fusion encoder (ASFE + RRM) that enhances RGB representation via noise-suppressed nDSM alignment and context refinement.
- 2) We design GCSD—a structure-preserving graph distillation module that performs graph reasoning on 2-D spatial patches, overcoming limitations of previous vectorized approaches.
- 3) We introduce an SPD module that injects semantic attention cues into the student decoder using teacher and student features, without relying on text prompts.
- 4) Our student model achieves a performance comparable to or superior to those of previous approaches on the Potsdam and Vaihingen datasets, with only 8.89 M parameters and 2.29 G floating-point operations (FLOPs).

II. RELATED WORK

A. Multimodal Remote Sensing Semantic Segmentation Networks

Owing to the inherent limitations of using a single data source, it is challenging to comprehensively characterize all the information contained within remote sensing imagery. With the rapid advancement of satellite sensing technologies, there has been a significant expansion in the availability of varied data sources, including hyperspectral imagery, digital terrain models, digital elevation models, and nDSM. These emerging data types provide complementary elevation and structural geometric features that are not present in standard RGB images, making multimodal fusion a critical research focus in the field of remote sensing semantic segmentation.

In recent years, dual-branch network architectures—where each modality independently extracts features before fusion—have been widely used for remote sensing semantic segmentation, effectively boosting segmentation performance

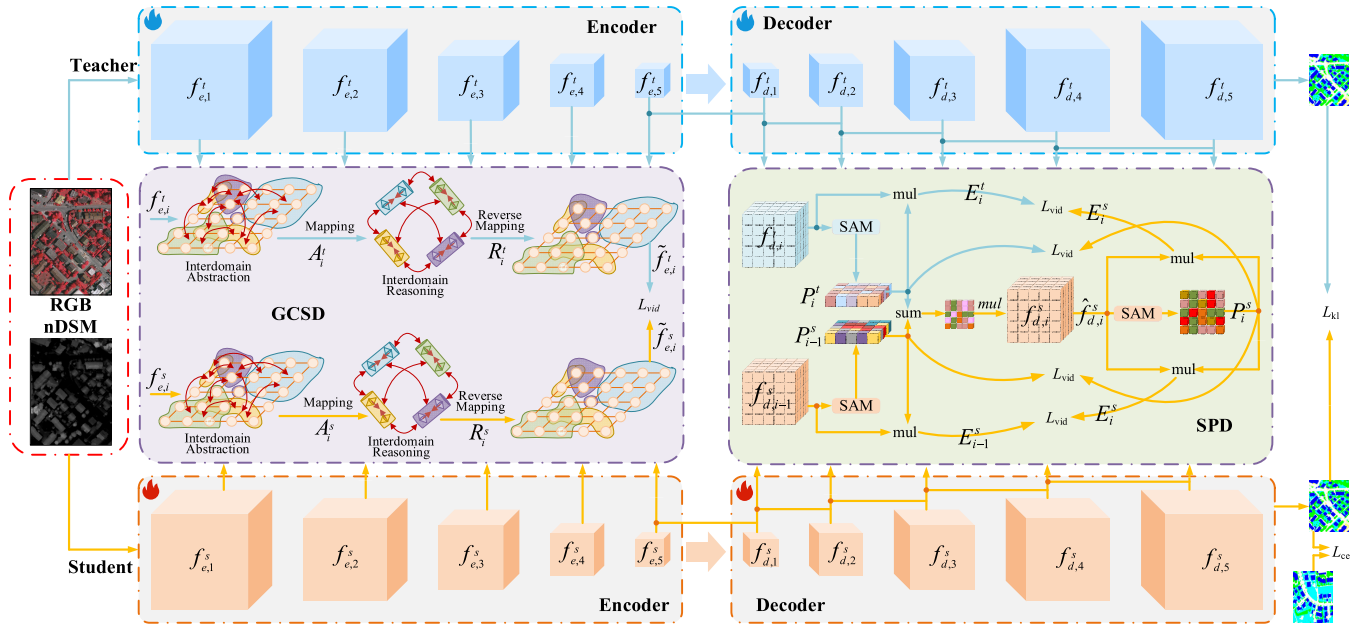


Fig. 1. Overall framework of SPGSNet with KD. ASFE: auxiliary spatial feature extraction; RRM: RGB representation module; GCSG: graph-convolution-structure distillation; SPD: semantic prompt distillation; SAM: semantic abstraction module. The SAM module integrates global contextual reasoning (via an affinity matrix) and local detail enhancement (via a dual-attention module) to extract semantic prompts from teacher/student features.

[22]. Recent dual-branch architectures employ three fusion strategies that are categorized based on fusion timing: early fusion [23], middle fusion [17], [24], and late fusion [25]. Although such approaches improve the accuracy of single-modality methods, they often overlook modality misalignment and nDSM noise, particularly in urban occlusions or dense vegetation areas. Attention mechanisms [26], dynamic weighting fusion [27], and cross-modal interaction modules [28] have been introduced to address these issues. However, noise and heterogeneity in elevation data—particularly in nDSM—remain critical bottlenecks in the fusion process, especially in areas with urban occlusions or dense vegetation coverage.

In summary, existing multimodal methods face persistent challenges in handling nDSM noise and modality heterogeneity. Although attention mechanisms [26] and dynamic fusion strategies [27] are effective in enhancing cross-modal interaction, they primarily rely on feature weighting and lack explicit denoising and spatial alignment mechanisms. By contrast, our ASFE module and RRM explicitly address nDSM noise and spatial misalignment through bilateral filtering, multi-scale recalibration, and graph-based context modeling, thereby providing a more robust and geometry-aware foundation for RGB–nDSM fusion.

B. Knowledge Distillation

With the growing demand for real-time performance and on-device deployment in remote sensing applications, lightweight semantic segmentation models have attracted increasing attention. However, deep learning models currently used in remote sensing semantic segmentation tasks often have large parameter counts and require complex computations, which hinder deployment on resource-constrained devices. KD is a widely adopted model compression technique that addresses this

limitation; it transfers knowledge from a high-capacity teacher network to a compact student network, enabling the construction of efficient yet accurate models—well aligned with the objectives of our research.

KD employs soft labels [18], intermediate features [29], or attention maps [30] to transfer knowledge from teacher to student networks. Advanced methods capture interlayer relations via FSP matrices [31] or cross-layer semantic alignment [32]; however, these methods were designed for classification and overlook spatial dependencies critical for segmentation. Recent graph-based approaches [33], [34], [35] attempt structural distillation by extracting multiscale cross-level information; nevertheless, they rely on simplifications such as global pooling or feature vectorization before graph reasoning. These operations disrupt spatial topologies—an essential characteristic for high-resolution remote sensing tasks—and focus only on encoder-level or channel-wise relations, overlooking spatial dependencies between nonadjacent pixels critical for dense prediction.

In summary, conventional KD methods [18], [29], [30] mainly focus on feature mimicry or soft-label transfer, whereas recent graph-based distillation approaches published in 2024–2025 [33], [34], [35] often compromise spatial topology owing to feature vectorization and graph construction on flattened representations. Unlike these methods, our GCSG module preserves spatial topology by performing graph reasoning directly on 2-D feature maps, maintaining the spatial continuity and structural consistency that are critical for dense predictions in remote sensing scenarios.

C. Prompt Learning

PL has emerged as a promising paradigm in deep learning. Its core idea is to guide the model to produce desired outputs by constructing suitable prompts, thereby enabling the efficient

utilization of pretrained models. Depending on the type of prompt features, PL methods can be divided into three main categories: textual, multimodal, and visual prompts.

PL was initially proposed in the natural language-processing domain through textual prompt methods [36] and has achieved remarkable success. It was later extended to vision-language tasks with the development of models such as CLIP [20] and CoOp [37]. For example, Khattak et al. [38], [39] introduced a multimodal PL strategy and a forget-free adaptation method for foundation models, treating prompts from both the image and text branches as equivalent and jointly learning them. Beyond vision-language applications, Jia et al. [19] further extended PL to the pure vision domain by proposing visual prompt tuning, which introduces learnable prompts in the input space to help pretrained models adapt more efficiently to downstream tasks. Building on this, Han et al. [40] proposed an efficient prompt-tuning approach that injects prompt tokens into the input space and embeds matrices into the self-attention mechanism while employing pruning strategies to reduce the number of prompts, thereby lowering the parameter count and computational overhead.

Overall, most existing PL methods rely heavily on paired textual labels or pretrained vision-language foundation models such as CLIP. However, such resources are rarely available in the field of remote sensing, where high-quality image-text annotations are scarce or nonexistent. Moreover, current prompt-based approaches are typically designed for classification or vision-language understanding tasks, rather than dense prediction scenarios such as semantic segmentation.

In summary, existing PL methods [19], [20], [21], [36], [37], [38], [39], [40] rely heavily on textual supervision or vision-language models such as CLIP, which are unavailable in remote sensing. Our SPD module generates visual semantic prompts dynamically from teacher features without textual dependency, filling the gap for dense prediction tasks under limited supervision.

III. PROPOSED METHOD

A. Overview

The proposed framework transfers knowledge from a high-performing teacher model, SPGSNet-T, to a compact student model, SPGSNet-S, resulting in an efficient student variant called SPGSNet-S*. Both SPGSNet-T and SPGSNet-S share the same architectural design; however, SPGSNet-T employs a ResNet50 [41] backbone for robust performance, whereas SPGSNet-S adopts a MobileNetV2 [42] backbone for lightweight efficiency.

During the encoding stage, the GCSD module extracts structural information from SPGSNet-T to guide structural feature learning in SPGSNet-S. In the decoding stage, the SPD module injects visual semantic prompt information from the teacher to the student model. This module also applies a self-distillation strategy, enabling the student model to independently explore and acquire additional knowledge. Finally, during the prediction stage, traditional response distillation supervises the student model using soft targets and label maps from the teacher model, further refining its performance.

B. SPGSNet

SPGSNet adopts a dual-stream encoder-decoder architecture, as shown in Fig. 2. Initially, two independent branches process RGB and nDSM data separately to extract distinct feature representations. The RGB features, R_i , and nDSM features, D_i , from corresponding layers are then input to the ASFE module. This module calibrates features and extracts spatial auxiliary information, producing calibrated nDSM features ($\tilde{d}_i$), calibrated RGB features (r_i), and spatial auxiliary features (w_i). Next, r_i , $\tilde{d}_i$, and w_i are input to the RRM, where w_i guides the fusion of r_i and $\tilde{d}_i$ to generate fused features, f_i ($i \in 1, 2, 3, 4, 5$), incorporating global contextual information. Finally, the decoding stage employs decoders to restore pixel-level information and upsample the fused features, f_i .

1) *Auxiliary Spatial Feature Extraction*: nDSM data obtained through existing technologies often contain significant noise. Aligning and denoising nDSM data with RGB features is challenging owing to differences in their imaging principles. To address these limitations, we first apply bilateral filtering to smooth and denoise the D_i features, producing $\tilde{D}_i$ features. Next, R_i and $\tilde{D}_i$ are sequentially fed into the multiscale spatial and channel self-calibration modules. These modules leverage respective strengths for mutual feature enhancement, generating aligned RGB and nDSM features (r_i and $\tilde{d}_i$, respectively), along with spatial auxiliary weights w_i . This process is illustrated in Fig. 3 and mathematically defined as follows:

$$BF(D_i) = \sum_{p \in D_i} e^{-\frac{\|p-q\|^2}{2\sigma_s^2}} * e^{-\frac{\|I_p - I_q\|^2}{2\sigma_r^2}} \quad (\sigma_s = 0.1, \sigma_r = 2.0) \quad (1)$$

$$\tilde{D}_i = BF(D_i) \quad (2)$$

$$\bar{R}_i = Cat(max(R_i) + mean(R_i)) \quad (3)$$

$$W_i^{sr} = CS_{k=7}(\bar{R}_i \times CS_{k=5}(\bar{R}_i \times CS_{k=3}(\bar{R}_i))) \quad (4)$$

$$D_i^s = W_i^{sr} \times \tilde{D}_i \quad (5)$$

$$\tilde{D}_i = Cat(max(D_i^s) + mean(D_i^s)) \quad (6)$$

$$W_i^{sd} = CS_{k=7}(D_i^s \times CS_{k=5}(D_i^s \times CS_{k=3}(D_i^s))) \quad (7)$$

$$R_i^s = W_i^{sd} \times R_i \quad (8)$$

$$W_i^{cr} = MS(MA_{s=5}(R_i^s \times MS(MA_{s=3}(R_i^s \times MS(MA_{s=1}(R_i^s)))))) \quad (9)$$

$$D_i^c = W_i^{cr} \times \tilde{D}_i \quad (10)$$

$$W_i^{cd} = MS(MA_{s=5}(D_i^c \times MS(MA_{s=3}(D_i^c \times MS(MA_{s=1}(D_i^c)))))) \quad (11)$$

$$r_i = R_i \times W_i^{cd} \times W_i^{cr} + R_i \quad (12)$$

$$\tilde{d}_i = \tilde{D}_i \times W_i^{cd} + \tilde{D}_i \quad (13)$$

where i ranges from 1 to 5. $BF(\cdot)$ represents bilateral filtering and $max(\cdot)$ and $mean(\cdot)$ indicate maximum and average pooling, respectively. $Cat(\cdot)$ denotes channel-wise feature concatenation and $CS(\cdot)$ represents feature convolution followed by $Sigmoid(\cdot)$ normalization, where k is the kernel size. $MA(\cdot)$ includes adaptive maximum and average pooling, with s denoting the pooling feature size, and $MS(\cdot)$ combines a multilayer perceptron and $Sigmoid(\cdot)$. The symbols $+$ and $\times$ represent element-wise addition and multiplication, respectively.

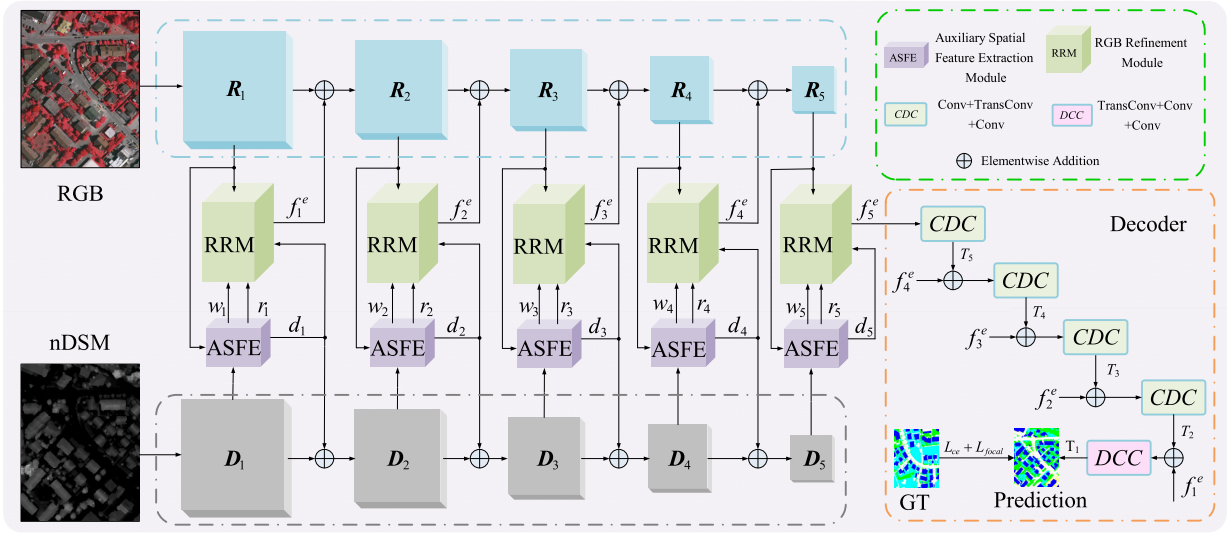


Fig. 2. Structure of the designed SPGSNet.

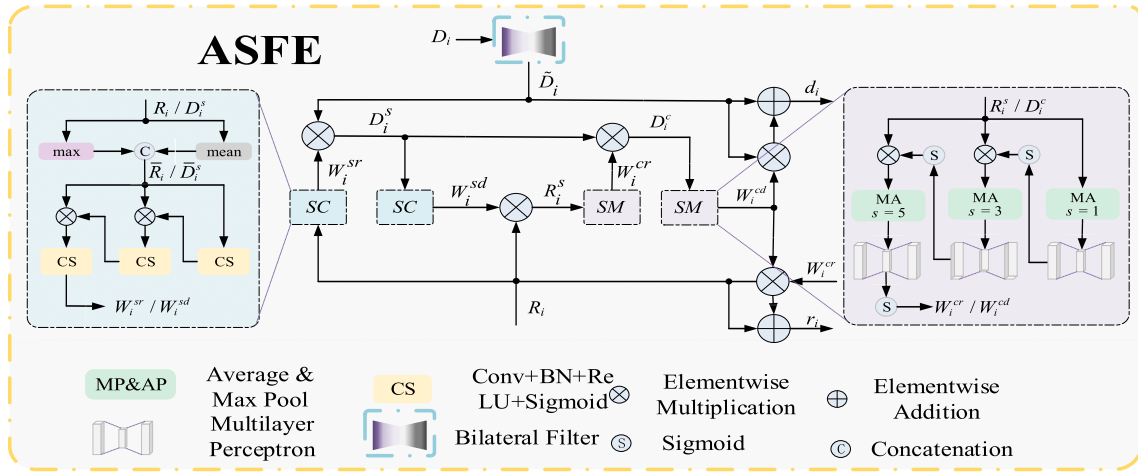


Fig. 3. Structure of the proposed ASFE module.

2) *RRM*: To effectively capture global semantic information in remote sensing images and address class imbalance, we designed the RRM. A literature review [43] revealed that many existing attention mechanisms reduce the spatial dimension to one for channel attention computation or reduce the channel dimension to two for spatial attention computation. Such reductions can result in considerable information loss due to the abrupt drop in spatial or channel dimensions. Inspired by spatial pyramid pooling in deep convolutional networks for visual recognition [44], we integrated a multiscale self-calibration module into the RRM. This module employs cascaded convolution blocks and multilayer perceptrons for multiscale spatial operations (blue in Fig. 4) and channel self-calibration (purple in Fig. 4).

The RRM consists of two branches: a global branch based on GCN [45] and a local branch using multiscale self-calibration. First, the outputs r_i and d_i from the ASFE module are concatenated along the channel dimension, and spatial guidance from w_i is injected to form H_i . Subsequently, H_i is fed into the global branch, which includes three cascaded GCNs to capture long-range dependencies across the image,

yielding G_i . To refine local pixel relationships, H_i is also input to the local branch, comprising spatial and channel self-calibration modules, producing refined features C_i and weights w_i^{ch} . To enhance attention to under-represented or previously overlooked classes, we introduce a reverse attention mechanism in which H_i is processed with w_i^{ch} and $1 - w_i^{ch}$, resulting in L_i and L_i^r . Finally, G_i , L_i , and L_i^r are sequentially fused through concatenation and convolution operations to yield refined RGB features, $\tilde{R}_i$. To mitigate information loss, $\tilde{R}_i$ is connected to r_i through a residual connection to obtain the final fused features, f_i . The mathematical formulation for this process is as follows:

$$H_i = (r_i + d_i) \times w_i \quad (14)$$

$$G_i = \text{Sigmoid}(\text{GCN}(\text{GCN}(\text{GCN}(H_i)))) \times H_i \quad (15)$$

$$C_i = H_i \times \text{CS}_{k=7}(H_i \times \text{CS}_{k=5}(H_i \times \text{CS}_{k=3}(\text{MA}(H_i)))) \quad (16)$$

$$w_i^{ch} = \text{MS}(\text{MA}_{s=5}(C_i \times \text{MS}(\text{MA}_{s=3}(C_i \times \text{MS}(\text{MA}_{s=1}(C_i)))))) \quad (17)$$

$$L_i = C_i \times w_i^{ch}, L_i^r = C_i \times (1 - w_i^{ch}) \quad (18)$$

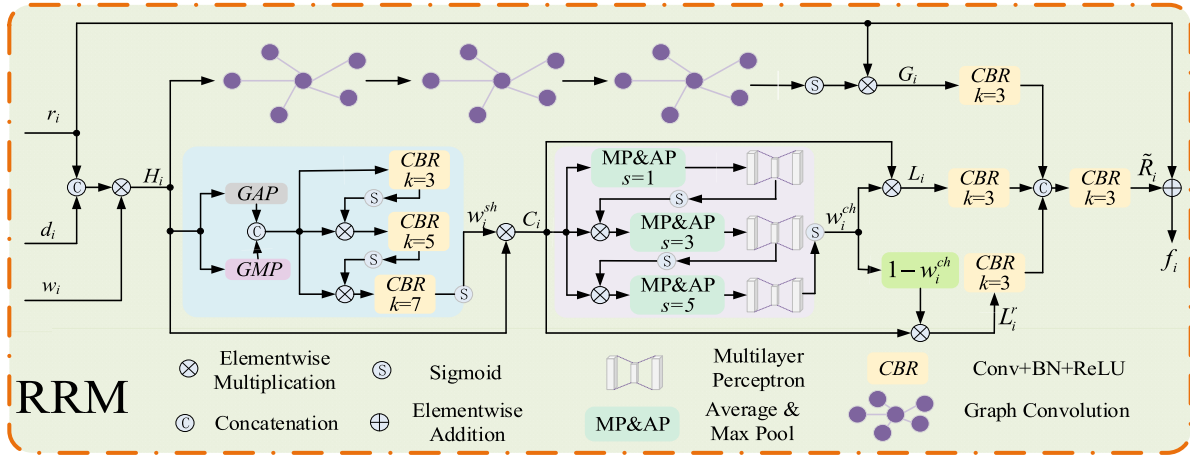


Fig. 4. Structure of the proposed RRM.

$$f_i = CBR(Cat(CBR(G_i), CBR(L_i), CBR(L'_i))) + r_i \quad (19)$$

where $i = 1, 2, 3, 4$, and 5 . $GCN(\cdot)$ represents graph convolution.

Finally, in the decoding stage, the fused features f_i output by the RRM are fed into adjacent decoders for pixel restoration and dimensionality reduction, producing the final prediction map. In this stage, we replace traditional upsampling with transposed convolutions. By leveraging learnable weights in transposed convolutions, the model acquires additional knowledge during decoding and reduces bias typically introduced by interpolation-based methods. The mathematical formulation of this process is as follows:

$$\begin{cases} T_1 = DCC(f_1^e + T_2), & i = 1 \\ T_i = CDC(f_i^e + T_{i+1}), & i = 2, 3, 4 \\ T_5 = CDC(f_5^e), & i = 5 \end{cases} \quad (20)$$

where T_1 – T_5 represent the decoded features at intermediate layers, with T_1 denoting the final prediction map; $CDC(\cdot)$ refers to a sequence of convolution, transposed convolution, and convolution operations; and $DCC(\cdot)$ denotes a sequence of transposed convolution, convolution, and convolution operations.

C. SPGSNet-KD

To construct a high-performance, yet compact model, we adopt a PL and KD strategy to transfer knowledge from the trained teacher model (SPGSNet-T) to the student model (SPGSNet-S). This approach enables the lightweight student to achieve a performance comparable to that of the teacher. Encoding features from shallow layers contain richer structural and detail-level information, whereas decoding features offer more abstract semantic representations. To exploit these complementary properties, we design two distinct distillation strategies: GCSD and SPD. GCSD transfers structural information from encoding features, whereas SPD distills semantic information from decoding features. The overall distillation framework is illustrated in Fig. 1 and discussed in detail in the following:

1) *GCSD*: Traditional graph-based distillation methods typically flatten 2-D feature maps into 1-D vectors, compromising spatial relationships that are crucial for semantic segmentation.

Our GCSD module addresses this limitation by operating directly on 2-D feature maps through three key steps: cross-domain abstraction, graph-based structural reasoning, and topology-preserving mapping. This design preserves spatial topologies to enable effective knowledge transfer in remote sensing scenes with irregular boundaries and diverse object scales. Specifically, GCSD extracts encoding features ($f_{e,i}^t$ and $f_{e,i}^s$) from the corresponding layers of the teacher and student models, respectively, and uses them as a pair of input features.

In the first step (cross-domain abstraction), features from different regions of $f_{e,i}^t$ and $f_{e,i}^s$ are abstracted to generate cross-domain abstract features A_i^t and A_i^s

$$A_i^t = IA(f_{e,i}^t), A_i^s = IA(f_{e,i}^s) \quad (21)$$

where $IA(\cdot)$ denotes the interdomain abstraction module, which extracts multiscale structural patterns through parallel convolution blocks with kernel sizes $\{1 \times 1, 3 \times 3, 5 \times 5\}$. Specifically

$$IA(F) = Cat(Conv_{1 \times 1}(F), Conv_{3 \times 3}(F), Conv_{5 \times 5}(F)) \quad (22)$$

where $Cat(\cdot)$ denotes channel-wise concatenation. This design enables the simultaneous capture of fine-grained edges (1×1), local structures (3×3), and coarse layouts (5×5), which is essential for modeling irregular boundaries in remote sensing imagery.

In the second step (graph-based structural reasoning), the abstracted features $A_i^t \in \mathbb{R}^{B \times C \times H \times W}$ and $A_i^s \in \mathbb{R}^{B \times C \times H \times W}$ are mapped to the graph space for topological modeling. Each spatial location in the 2-D feature map becomes a vertex with its channel-wise features as node attributes. We adopt eight-neighborhood connectivity to capture diagonal relationships prevalent in remote sensing imagery, particularly for preserving boundaries of oblique structures such as buildings and roads. The mapping is formulated as

$$G_i^t = M(A_i^t), G_i^s = M(A_i^s) \quad (23)$$

where $M(\cdot)$ transforms 2-D features into graph nodes while preserving spatial locality

$$M(F) = Reshape(F, [H \times W, C] + PE(H, W)) \quad (24)$$

where $PE(H, W)$ denotes 2-D sinusoidal positional encoding that encodes the original (x, y) coordinates. This ensures that graph reasoning can leverage the spatial proximity despite the node representation.

Subsequently, structural reasoning is performed through the Chebyshev GCN

$$R_i^t = GCN(G_i^t), \quad R_i^s = GCN(G_i^s). \quad (25)$$

We adopt Chebyshev GCN because it approximates spectral convolutions using K-order polynomials

$$GCN(H^{(l)}) = \sum_{k=0}^K \theta_k T_k(L) H^{(l)} \quad (26)$$

where $H^{(l)}$ is the feature matrix at layer l , T_k is the k th Chebyshev polynomial, $L = 2L/\lambda_{max} - I$ is the scaled Laplacian matrix, and θ_k are learnable parameters. With $K = 3$, each node aggregates information from neighbors up to three hops away, effectively capturing long-range spatial dependencies without dense connectivity, which is critical for modeling complex structures across multiple scales.

In the third step (topology-preserving mapping), the reasoning features, R_i^t and R_i^s , are mapped back to the 2-D space while preserving the structural enhancements

$$\hat{f}_i^t = RM(R_i^t), \quad \hat{f}_i^s = RM(R_i^s) \quad (27)$$

where $RM(\cdot)$ reconstructs spatial layouts from graph representations

$$RM(H) = Reshape(H - PE(H, W), [H, W, C]). \quad (28)$$

Here, we subtract the positional encoding before reshaping to avoid information redundancy. This inverse operation ensures minimal information loss while recovering the original spatial structure enhanced by graph reasoning.

Finally, the variational information distillation (VID) loss is used to measure the discrepancy between $\hat{f}_i^t$ and $\hat{f}_i^s$

$$VID(\hat{f}_i^t, \hat{f}_i^s) = \frac{1}{N} \sum_{j=1}^N \left[\log(\alpha_j + \epsilon) + \frac{(\hat{f}_j^t - \hat{f}_j^s)^2}{2\alpha_j} \right] \quad (29)$$

$$L_{GSD} = \sum_{i=1}^5 l_{vid}(\hat{f}_i^t, \hat{f}_i^s) \quad (30)$$

where α is a learnable parameter that adapts to the uncertainty in different structural patterns and ϵ is a small constant added for numerical stability.

2) *Semantic Prompt Distillation*: Existing distillation methods typically treat teacher features as mere supervisory signals, failing to fully leverage their semantic content. Our SPD module introduces a visual prompt distillation paradigm that dynamically generates semantic prompts from teacher features to guide student learning. At the core of SPD is the semantic abstraction module (SAM), which extracts high-level semantic information through a unified three-step process: global contextual reasoning, local detail enhancement, and adaptive integration. This design eliminates text dependency while providing rich contextual guidance for dense prediction tasks.

Specifically, SPD takes as input the student feature $f_{d,i}^s$, teacher feature $f_{d,i}^t$, and low-level student feature f_{i-1}^s . Notably, in the initial decoding stage, the low-level student feature is replaced by the final-layer feature $f_{e,5}^s$ from the encoding stage. These features are processed through $SAM(\cdot)$ to generate semantic prompt features

$$P_i^t = SAM(f_{d,i}^t) \quad (i = 1, 2, 3, 4, 5) \quad (31)$$

$$\begin{cases} P_1^s = SAM(f_{e,5}^s) \\ P_i^s = SAM(f_{d,i}^s) \end{cases} \quad (i = 2, 3, 4, 5). \quad (32)$$

In the first step (global contextual reasoning), SAM computes an affinity matrix $A \in \mathbb{R}^{N \times N}$ to model semantic relationships between all spatial locations

$$A = softmax(F_\theta F_\varphi^T) \quad (33)$$

where $F_\theta = Conv_{1 \times 1}(F)$ and $F_\varphi = F^T$. This affinity matrix is particularly effective in remote sensing imagery because it captures relationships between spatially distant but semantically similar regions, such as disconnected areas of the same land cover type. The Softmax operation normalizes these relationships, emphasizing the most relevant connections. The global context-aware features are obtained by

$$F_{global} = A \times F. \quad (34)$$

In the second step (local detail enhancement), SAM applies dual attention mechanisms to capture complementary channel-wise and spatial dependencies. Channel attention identifies “what” is semantically important by modeling interchannel relationships

$$C_{channel} = Sigmoid(MLP(Cat(GAP(F), GMP(F)))) \quad (35)$$

where GAP and GMP denote global average and max-pooling across spatial dimensions, respectively, and MLP is a two-layer perceptron with ReLU activation. Spatial attention identifies “where” to focus by modeling interspatial relationships

$$C_{spatial} = Sigmoid(Conv_{7 \times 7}(Cat(max(F), mean(F)))) \quad (36)$$

where $max(\cdot)$ and $mean(\cdot)$ operate along the channel dimension to generate two spatial context descriptors, and $Conv_{7 \times 7}$ is a 7×7 convolution for capturing local spatial patterns. This dual attention mechanism addresses a key challenge in remote sensing segmentation—the need to simultaneously capture global context (via channel attention focusing on semantic categories) and local details (via spatial attention preserving object boundaries). By combining maximum and average operations, the model learns the most salient features as well as the general feature distribution. The locally enhanced features are

$$F_{local} = F \times C_{channel} \times C_{spatial}. \quad (37)$$

In the third step (adaptive integration), the global and local branches are fused to balance semantic context and fine-grained details

$$SAM(F) = \lambda \cdot F_{global} + (1 - \lambda) F_{local} \quad (38)$$

where λ is a learnable parameter initialized to 0.5, enabling the model to automatically adjust the tradeoff between global semantic reasoning and local detail preservation based on

the characteristics of different feature levels. This integrated design ensures that SAM can adapt to varying requirements across the decoding hierarchy—deeper layers benefit more from global context, whereas shallower layers require stronger local details.

Subsequently, the semantic prompts P_i^t and P_{i-1}^s are accumulated, fused, and injected into student features $f_{d,i}^s$ to obtain the initial semantic-enhanced feature $\hat{f}_{d,i}^s$

$$\hat{f}_{d,i}^s = f_{d,i}^s \times (P_i^t + P_{i-1}^s). \quad (39)$$

This fusion operation leverages both teacher knowledge (from P_i^t) and the student's own representation (from P_{i-1}^s), creating a semantic guidance signal that is more comprehensive than that achieved by either source alone. The multiplicative integration ensures that the prompts modulate the original features without overwhelming them, maintaining a balance between external guidance and internal learning. This mechanism guides the student model to focus on semantically meaningful regions by highlighting class-specific features derived from the teacher model.

Next, to enable multilevel semantic abstraction, $SAM(\cdot)$ is applied recursively to $\hat{f}_{d,i}^s$

$$P_i^s = SAM(\hat{f}_{d,i}^s). \quad (40)$$

This recursive application captures hierarchical semantic relationships common in remote sensing imagery, such as buildings within urban areas or diverse vegetation types within forests. Each application of SAM refines the semantic representation at a higher level of abstraction.

Finally, the input features $f_{d,i}^t$, $f_{d,i}^s$, and $f_{d,i-1}^s$ are multiplied element-wise with the corresponding abstract semantic features P_i^t , P_i^s , and P_{i-1}^s to obtain the final semantically enhanced features E_i^t , E_i^s , and E_{i-1}^s

$$E_i^t = f_{d,i}^t \times P_i^t, E_i^s = f_{d,i}^s \times P_i^s, E_{i-1}^s = f_{d,i-1}^s \times P_{i-1}^s. \quad (41)$$

This element-wise multiplication acts as an attention mechanism, directing the model to the most semantically relevant regions of the feature maps. In remote sensing, this helps mitigate class imbalance by emphasizing less frequent but critical classes (e.g., cars or small structures) that might otherwise be overshadowed by dominant categories such as vegetation or buildings.

The VID loss measures the discrepancy between the abstract semantic features (P_i^t , P_i^s , and P_{i-1}^s) and the final semantic enhanced features (E_i^t , E_i^s , and E_{i-1}^s)

$$L_{SPD} = l_{vid}(P_i^t, P_i^s) + l_{vid}(P_i^s, P_{i-1}^s) + l_{vid}(E_i^t, E_i^s) + l_{vid}(E_i^s, E_{i-1}^s). \quad (42)$$

This multiterm loss provides a comprehensive supervisory signal, aligning the generated prompt maps and resulting enhanced features. It ensures that the semantic attention mechanism and its influence are consistently transferred from the teacher to the student, enabling the student to learn not only what features to extract but also how to semantically prioritize them.

The complementary nature of GCSD and SPD forms a holistic KD framework: while GCSD transfers structural knowledge during encoding, capturing spatial relationships and object

boundaries, SPD distills semantic knowledge during decoding, emphasizing class-specific features and abstract category representations. Together, they encompass the full spectrum of information needed for accurate remote sensing image segmentation.

D. Total Loss

When training SPGSNet-T, we use the ground-truth label y to supervise the predicted maps $\hat{y}$. Initially, the traditional cross-entropy loss is applied to measure the overall difference between y and $\hat{y}$. Next, focal loss is employed to further supervise the predictions, addressing class imbalance and improving the model's focus on hard-to-classify samples. The corresponding formulas are as follows:

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N y_i * \log \hat{y}_i \quad (43)$$

$$L_{Focal} = -\frac{1}{N} \sum_{i=0}^N \alpha * (1 - \hat{y}_i)^\gamma * \log \hat{y}_i (\gamma = 2) \quad (44)$$

$$L_T^{total} = L_{CE} + L_{Focal}. \quad (45)$$

During the distillation process, in addition to employing GCSD and SPD strategies to transfer knowledge from the teacher to the student model, we incorporate traditional response distillation in the prediction phase. This process uses both cross-entropy loss and KL divergence to refine the student model's output. The overall distillation loss L_{KD} is formulated as

$$KL(\hat{y}_i^t || \hat{y}_i^s) = \sum_{i=1}^{w*h} \hat{y}_i^t * \log \frac{\hat{y}_i^t}{\hat{y}_i^s} \quad (46)$$

$$L_{Response} = KL(\hat{y}_i^t, \hat{y}_i^s) + L_{CE}(\hat{y}_i^s, y) \quad (47)$$

$$L_{KD} = \alpha (L_{GCSD} + L_{SPD}) + (1 - \alpha) L_{Response} \quad (48)$$

where y , $\hat{y}_i^s$, and $\hat{y}_i^t$ denote the ground-truth label map, student prediction map, and teacher prediction map, respectively, and α is a hyperparameter set to 0.3.

IV. EXPERIMENTS AND RESULTS

A. Datasets and Evaluation Metrics

To verify the performance of the proposed SPGSNet-S* (SPGSNet with KD), we conducted experiments on two large-scale datasets: Potsdam and Vaihingen. These datasets contain six categories: low vegetation, impervious surfaces, buildings, cars, trees, and clutter/background. In addition, we explored RGB-D mirror segmentation [44] using an indoor dataset to further validate the robustness of the model.

1) *Vaihingen and Potsdam Datasets*: The Vaihingen dataset comprises 33 RGB images with a size of 2500×2000 along with the corresponding nDSM data. Following the partitioning protocol commonly adopted in recent literature [16], [51], [53], we selected 11 images (IDs: 1, 3, 5, 7, 13, 17, 21, 23, 28, 33, 37) for training, five images (IDs: 11, 15, 26, 30, 34) for validation, and the remaining 17 images for testing. The Potsdam dataset contains 38 RGB images with a size of 6000×6000 with the corresponding nDSM data; we randomly selected 24 images for training, seven for validation, and seven

TABLE I
QUANTITATIVE COMPARISON OF THE VAIHINGEN DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

	Imp. surf		Building		Low veg		Tree		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S ₂₀₁₅ [4]	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
SegNet ₂₀₁₇ [5]	89.88	80.93	90.88	86.54	78.66	64.07	88.96	75.96	43.93	43.16	78.46	70.13
DeepLabv3+ ₂₀₁₈ [6]	90.06	81.11	87.04	82.7	76.65	64.44	88.60	76.64	42.51	43.10	76.97	69.49
ACNet ₂₀₁₉ [7]	91.95	85.34	95.45	91.82	78.64	66.87	91.20	78.55	83.12	76.81	88.07	79.88
SA-Gate ₂₀₂₀ [8]	90.99	85.70	93.85	91.72	84.95	68.68	89.06	79.15	84.27	78.07	88.62	80.67
HRCNet ₂₀₂₁ [46]	91.62	81.60	91.72	88.01	79.24	67.38	90.55	78.58	70.69	68.73	84.76	76.86
ESANet ₂₀₂₁ [47]	92.09	85.18	94.93	91.16	75.72	65.48	92.35	77.65	75.92	70.11	86.20	77.92
CMXNet ₂₀₂₃ [12]	91.78	85.39	95.15	91.28	77.00	66.29	92.59	78.85	83.23	78.58	87.95	80.08
MGCNet ₂₀₂₃ [48]	91.18	84.89	95.58	91.52	76.22	65.74	92.76	78.53	79.56	75.33	87.06	79.20
SFAF-MA ₂₀₂₃ [50]	90.94	85.37	95.67	92.21	80.99	67.55	90.78	78.95	74.74	71.19	86.62	79.05
EAEFNet ₂₀₂₃ [49]	93.78	84.37	95.35	92.25	79.87	67.98	89.11	79.00	81.54	74.24	87.93	79.57
CATNet ₂₀₂₄ [13]	90.53	83.98	95.27	91.08	78.27	65.62	90.57	77.72	70.28	61.85	84.98	76.05
CAINet ₂₀₂₄ [17]	91.29	84.89	95.65	92.21	78.61	66.85	91.55	78.67	82.60	78.15	87.94	80.15
STRD-Net ₂₀₂₄ [16]	91.03	83.30	94.26	90.42	81.06	67.20	89.49	78.57	44.86	43.57	80.14	72.61
MSGCNet ₂₀₂₄ [53]	91.20	84.92	94.93	90.94	77.73	67.04	92.34	78.19	85.06	78.69	88.25	79.95
AFENet ₂₀₂₅ [51]	91.88	84.52	93.96	91.02	80.29	67.66	91.03	78.59	82.94	78.92	88.02	80.14
CPSSNet ₂₀₂₅ [52]	93.28	83.28	95.31	90.71	71.28	64.44	94.02	79.04	83.05	77.26	87.39	78.95
SDGCNet-T	93.35	85.19	95.45	92.40	78.20	67.56	91.30	79.20	91.03	85.16	89.87	81.90
SDGCNet-S	91.91	85.27	95.63	92.25	75.86	66.39	93.17	78.17	75.99	70.77	86.50	78.57
SDGCNet-S*	91.84	85.98	96.26	92.73	79.11	67.83	91.82	79.15	85.86	78.91	88.92	80.92

for testing, following the split ratios used in [51]. Given the large size of the original images, all images were cropped into 256×256 patches with 50% overlap during training and nonoverlapping patches during testing, following the standard practice in high-resolution remote sensing segmentation [16], [51], [53]. This preprocessing ensures computational efficiency while preserving spatial context for accurate boundary delineation.

2) *Evaluation Metrics*: To evaluate the performance of SPGSNet-S*, we used the following metrics: intersection over union (IoU), mean IoU (mIoU), class accuracy (Acc), and mean accuracy (mAcc). Additionally, we compared the FLOPs and number of parameters (Params) to assess the lightweight design and computational efficiency of SPGSNet-S*. We conducted 200 iterations with batch sizes of 10 and 20, whereas for the student model training, we used 200 iterations with a batch size of 20.

B. Experimental Setup

We optimized the network using the Adam optimizer, with the initial parameters set as follows: the momentum was initialized at 0.9, the learning rate at $1e-4$, and weight decay at $1e-4$. During training, the momentum parameter was reduced to one-tenth of its current value every 20 epochs.

1) *Baseline Reproduction*: For fair comparison, all baseline methods (FCN-8S, SegNet, ACNet, CMXNet, MGCNet, STRD-Net, AFENet, etc.) were reimplemented using our dataset splits and training protocols. We strictly followed the architectural designs and hyperparameters reported in their original papers, with adjustments only to accommodate our data preprocessing (256×256 patches). Each baseline was trained for 200 epochs using the same hardware [NVIDIA TITAN XP GPU (12 GB)] and software (PyTorch 2.1.0) environments as those in our method. The results presented in Tables I and II represent the best performance achieved by each method under this unified experimental setting; these

results may differ from those in their original papers owing to differences in dataset splits, preprocessing strategies, or implementation details.

C. Results and Analysis

We compared our proposed SPGSNet-S* with various state-of-the-art methods on the Vaihingen and Potsdam datasets. The specific comparison results are presented in Tables I and II.

1) *Quantitative Evaluation*: By comparing the evaluation metrics of SPGSNet-T, SPGSNet-S, and SPGSNet-S* with other models (Tables I and II), we observed that SPGSNet-T outperformed most of the baseline methods in prediction accuracy across several individual categories, achieving the best results in mIoU and mAcc. Furthermore, based on our KD strategy, SPGSNet-S* achieved the second-best results in mIoU and mAcc. Although SPGSNet-S* approached but did not surpass SPGSNet-T in individual category prediction accuracy, the comparison with SPGSNet-S indicates that our KD strategy significantly improved performance across all categories. These results fully demonstrate the stability and effectiveness of our KD strategy. Specifically, the improvements of SPGSNet-S after distillation are as follows: mAcc and mIoU increased by 2.42% and 2.35%, respectively, on the Vaihingen dataset, and by 2.26% and 2.14%, respectively, on the Potsdam dataset.

2) *Qualitative Evaluation*: Fig. 5 presents the results of SPGSNet-S* compared with the most representative semantic segmentation models currently available. We selected three datasets from Potsdam and five from Vaihingen for comparison. The prediction maps clearly demonstrate that our model achieved excellent segmentation accuracy for both small, sparsely distributed objects and large, continuously distributed ones. This improvement is attributed to the ASFE module and RRM, which effectively capture elevation information from the nDSM and global contextual relationships from the RGB images. Furthermore, the student model SPGSNet-S*,

TABLE II
QUANTITATIVE COMPARISON OF THE POTSDAM DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

	Imp. surf		Building		Low veg		Tree		Car		Clutter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S ₂₀₁₅ [4]	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
SegNet ₂₀₁₇ [5]	90.18	80.46	90.21	84.18	85.88	71.63	82.49	70.68	93.15	89.72	51.76	37.21	82.28	72.31
DeepLabv3+ ₂₀₁₈ [6]	91.57	82.49	91.78	87.59	87.36	73.63	85.45	73.32	93.89	90.04	53.80	43.54	83.97	75.10
ACNet ₂₀₁₉ [7]	91.32	82.74	93.83	90.06	86.16	73.53	86.03	72.87	93.79	90.43	54.51	41.65	84.27	75.21
SA-Gate ₂₀₂₀ [8]	85.84	80.64	93.65	88.51	86.46	72.71	85.70	72.89	92.18	89.39	62.70	40.59	84.42	74.12
HRCNet ₂₀₂₁ [46]	90.03	81.68	90.87	85.75	88.17	73.18	82.02	71.32	93.94	89.82	56.72	40.03	83.63	73.63
ESANet ₂₀₂₁ [47]	91.38	82.92	93.69	89.82	87.10	73.16	82.48	70.81	93.08	88.53	55.68	43.38	83.90	74.77
CMXNet ₂₀₂₃ [12]	90.54	81.86	93.80	89.28	87.17	73.11	83.52	71.45	92.23	87.99	51.61	40.58	83.14	74.04
MGCNet ₂₀₂₂ [48]	90.61	81.21	93.87	89.02	86.72	72.67	83.70	72.51	93.39	88.91	53.24	39.66	83.59	74.00
SFAF-MA ₂₀₂₃ [50]	90.63	82.06	93.31	89.67	85.76	72.35	83.51	70.95	92.55	88.34	60.03	42.40	84.30	74.30
EAEFNet ₂₀₂₃ [49]	92.32	82.04	93.22	89.54	84.34	72.37	85.02	71.91	94.64	88.30	53.55	39.34	83.85	73.92
CATNet ₂₀₂₄ [13]	88.12	79.80	93.32	89.52	85.91	70.66	78.78	68.85	91.56	84.99	55.58	40.41	82.21	71.87
CAINet ₂₀₂₄ [17]	88.77	81.25	92.21	88.70	88.50	71.80	81.55	71.17	94.33	89.65	57.05	42.95	83.74	74.25
STRD-Net ₂₀₂₄ [16]	91.61	79.79	90.70	87.32	88.27	71.66	77.76	69.14	90.00	84.81	48.13	36.58	81.08	71.55
MSGCNet ₂₀₂₄ [53]	86.96	78.25	91.49	82.45	82.71	70.86	82.96	67.07	95.88	76.30	56.39	39.81	82.73	69.46
AFENet ₂₀₂₅ [51]	91.52	82.18	90.93	86.52	87.26	73.63	83.87	72.97	94.53	90.89	56.84	42.08	84.16	74.71
CPSSNet ₂₀₂₅ [52]	88.87	81.23	91.82	85.61	89.02	73.18	81.38	71.95	93.23	89.56	54.76	38.48	83.18	73.34
SDGCNet-T	91.47	82.27	93.93	89.61	87.19	73.96	86.16	74.36	95.26	90.88	55.41	44.23	84.91	75.89
SDGCNet-S	89.94	80.79	90.35	87.23	87.06	71.73	83.61	72.02	91.70	88.40	53.21	38.94	82.63	73.18
SDGCNet-S*	90.72	82.56	93.66	89.75	86.42	73.44	85.37	73.94	93.95	89.73	59.22	45.42	84.89	75.32

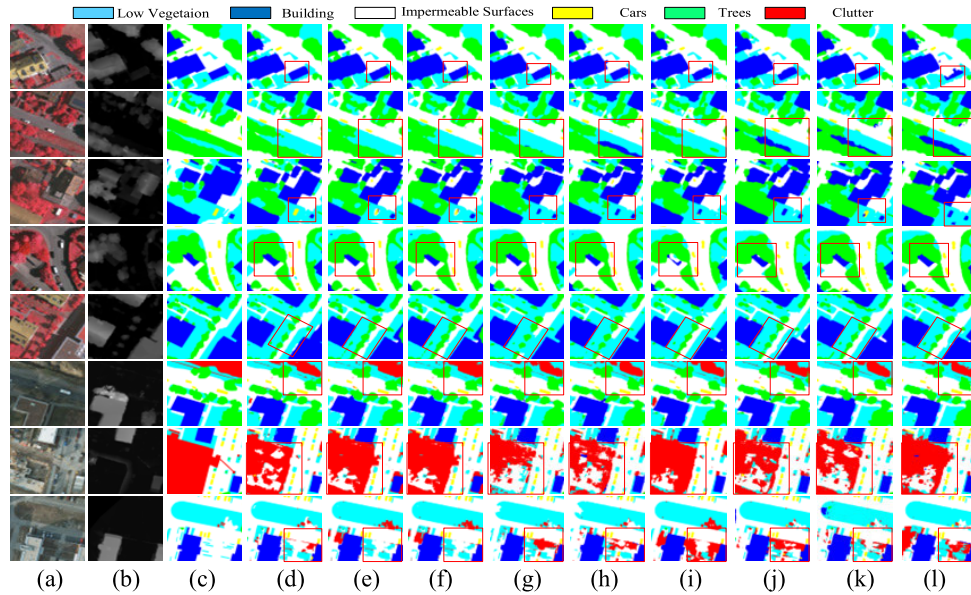


Fig. 5. Semantic segmentation results. (a) RGB. (b) nDSM. (c) GT. (d) SPGSNet-T. (e) SPGSNet-S. (f) SPGSNet-S*. (g) SFAF-MA. (h) MSGCNet. (i) EAEFNet. (j) AFENet. (k) MGCNet. (l) CMXNet. For ease of observation, we use red boxes to indicate the salient areas of the predictions.

after distillation with our proposed strategy, exhibited a performance that was comparable to that of the teacher model. This further validates the effectiveness of our proposed distillation strategy.

3) *Complexity Comparison*: Table III presents a comparison of our teacher and student models with other state-of-the-art networks in terms of parameter count and computational complexity. The results indicate that our proposed distillation strategy effectively transferred knowledge from the teacher model to the student model, enabling the student model to achieve competitive performance with only 8.89 M parameters and 2.29 G computations. Notably, SPGSNet-S* attained an mIoU comparable to those of recent dual-encoder methods, such as CMXNet, MGCNet, and TCNet, while requiring

significantly fewer computations (2.29 G versus 14.26 G FLOPs for CMXNet), thereby validating the effectiveness of our distillation-based compression strategy.

4) *F1-Score Analysis for Class Imbalance Assessment*: To provide a more comprehensive evaluation of our method's performance in class-imbalanced scenarios, we further analyzed the results using *F1*-scores, particularly focusing on underrepresented classes. As shown in Table IV, SPGSNet-S* demonstrated significant improvements in *F1*-scores across all categories, with particularly notable gains in small classes, such as the Car and Clutter/Background categories in the Vaihingen dataset. Specifically, the macroaveraged *F1*-score improved from 87.38% (SPGSNet-S) to 89.14% (SPGSNet-S*), whereas the average *F1*-score for small classes

TABLE III
COMPUTATIONAL COMPLEXITY COMPARISON. THE BEST
RESULT IS HIGHLIGHTED IN RED

Model	Backbone	Params (M)↓	Flops (G) ↓
FCN-8S ₂₀₁₅	VGG-16	134.29	74.55
SegNet ₂₀₁₇	VGG-16	53.56	18.3
ACNet ₂₀₁₉	ResNet-50	116.6	26.41
SA-Gate ₂₀₂₀	ResNet-101	110.85	41.23
HRCNet ₂₀₂₁	ResNet-50	62.71	30.27
ESANet ₂₀₂₁	ResNet-34	45.42	10.15
CMXNet ₂₀₂₂	SegFormer	66.56	14.26
MGCNet ₂₀₂₂	ConvNeXt	209.99	47.51
SFAF-MA ₂₀₂₃	ResNet-50	232.08	18.24
EAEFNet ₂₀₂₃	ResNet-50	77.97	23.3
CATNet ₂₀₂₄	Swim-Former	262.74	76.79
CAINet ₂₀₂₄	MobileNet-V2	13.8	4.69
STRD-Net ₂₀₂₄	ResNet-50	201.31	31.68
MSGCNet ₂₀₂₄	ResNet-34	27.61	14.25
AFENet ₂₀₂₅	ResNet-18	20.24	6.41
CPSSNet ₂₀₂₅	ConvNeXt	31.86	8.25
SPGSNet-T	ResNet-50	124.08	20.58
SPGSNet-S	MobileNet-V2	8.89	2.29
SPGSNet-S*	MobileNet-V2	8.89	2.29

increased from 89.71% to 90.51%. These results demonstrate that our KD strategy not only maintains strong performance on dominant classes but also effectively addresses the class imbalance issue by substantially improving the recognition accuracy of minority classes.

D. Ablation Studies

To systematically validate the effectiveness of our three main contributions, we conducted comprehensive ablation experiments on the Potsdam and Vaihingen datasets. The experimental design followed a clear logic. First, the multimodal feature enhancement modules (ASFE module and RRM, corresponding to Contribution 1) were evaluated, then the structure-preserving distillation strategy (GCSD, corresponding to Contribution 2) was assessed, and finally, the SPD mechanism (SPD, corresponding to Contribution 3) was examined. The quantitative results of these ablation studies are summarized in Tables V and VI, with qualitative comparisons provided in Figs. 6–8.

- 1) *Effectiveness of Multimodal Feature Enhancement (Validates Contribution 1)*: To validate Contribution 1, which involves the design of the ASFE module and RRM for noise-robust spatial alignment and RGB feature enhancement, we performed ablation experiments to assess their individual and combined effectiveness. Specifically, we compared: a) a baseline model without any fusion module; b) the ASFE module applied for modality alignment; c) the RRM applied for RGB context optimization; and d) the complete ASFE + RRM pipeline. In addition, to provide a fair comparison with commonly used fusion strategies, we further evaluated several standard cross-modal fusion baselines, including element-wise summation, concatenation followed by a 1×1 convolution, and gate-based fusion. These experiments were conducted using noisy or misaligned nDSM inputs to simulate realistic remote sensing scenarios. As presented in Table V and Fig. 6,

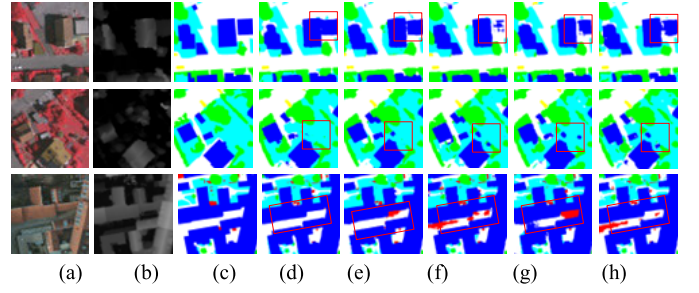


Fig. 6. Dense prediction comparison. (a) RGB. (b) nDSM. (c) GT. (d) SPGSNet-T. (e) SPGSNet-S*. (f) baseline. (g) WO/RRM. (h) WO/ASFE.

TABLE IV
PER-CLASS PERFORMANCE EVALUATION WITH F1-SCORES

Model	Imp. surf	Building	Low veg	Tree	Car	mF1
	F1	F1	F1	F1	F1	
STRD-Net ₂₀₂₄	84.03	84.14	78.72	84.01	83.62	88.82
MSGCNet ₂₀₂₄	89.83	89.46	84.32	88.71	90.43	88.55
AFENet ₂₀₂₅	90.03	89.27	84.23	89.68	90.16	88.67
CPSSNet ₂₀₂₅	89.33	89.22	82.22	88.04	90.40	87.84
SDGCNet-T	90.71	89.78	86.19	90.49	91.49	89.73
SDGCNet-S	87.66	88.72	83.58	87.22	89.71	87.38
SDGCNet-S*	89.92	89.30	85.36	90.59	90.51	89.14

TABLE V
ABLATION EXPERIMENT OF MODULES. THE BEST
RESULT IS HIGHLIGHTED IN RED

Modules	RRM	ASFE	VAIHINGEN		POTSDAM	
			mAcc	mIoU	mAcc	mIoU
Baseline (only sum)			85.12	76.34	81.45	71.02
Concat with 1×1 Conv			85.33	76.19	81.69	71.23
Gate fusion			85.78	77.02	82.03	72.45
WO/RRM		✓	87.03	78.86	83.86	73.21
WO/ASFE	✓		87.20	79.42	84.00	74.08
SPGSNet-T	✓	✓	89.87	81.90	84.91	75.89

the baseline model (single RGB branch without nDSM) achieved an mIoU of 76.34% on Vaihingen and 71.02% on Potsdam. Conventional fusion schemes, such as summation and concatenation, provide limited performance improvements owing to the lack of explicit noise suppression and spatial alignment. Although gate-based fusion introduces adaptive weighting and achieves better results than simple aggregation, it still underperforms the proposed approach. Both the ASFE module and RRM contributed positively to segmentation accuracy, and their combination consistently yielded the best performance across both datasets, confirming their complementary roles in handling modality noise and enhancing contextual features.

- 2) To validate Contribution 2, we assessed the performance of the proposed GCSD module by comparing our GCSD-based framework against conventional structure distillation methods such as SKD [34] and FSP [31]. The goal was to verify whether GCSD can better preserve spatial dependencies and region-level structural relationships between teacher and student representations. We also conducted a “W/O GCSD” variant to quantify the specific gains attributed to our structural distillation scheme.

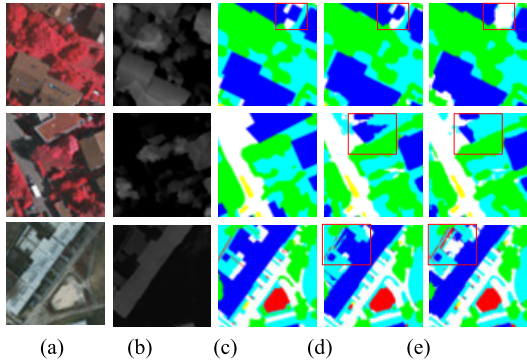


Fig. 7. Dense prediction comparison. (a) RGB. (b) nDSM. (c) GT. (d) SPGSNet-S*. (e) WO/GCSD.

TABLE VI

ABLATION EXPERIMENT OF THE DISTILLATION METHODS. THE BEST RESULT IS HIGHLIGHTED IN RED

Method	GCSD	SPD	VAIHINGEN		POTSDAM	
			mAcc	mIoU	mAcc	mIoU
SPGSNet-S			86.50	78.57	82.63	73.18
WO/GCSD		✓	87.93	79.86	83.97	74.33
WO/SPD	✓		87.76	79.48	83.93	74.16
SPGSNet-S*	✓	✓	88.92	80.92	84.89	75.32
SKD [34]			87.26	79.24	83.24	73.12
FSP [31]			87.33	79.29	83.05	72.91

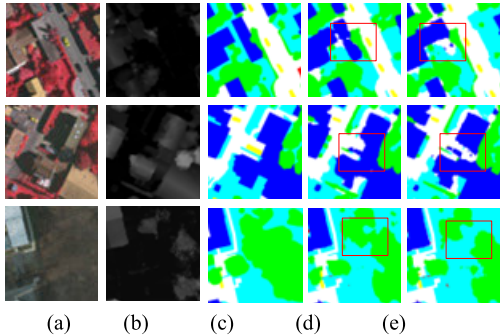


Fig. 8. Dense prediction comparison. (a) RGB. (b) nDSM. (c) GT. (d) SPGSNet-S*. (e) WO/SPD.

As shown in Table VI and Fig. 7, GCSD outperformed popular structural distillation methods such as SKD and FSP, confirming its ability to preserve spatial topology and guide region-level representation learning in dense prediction tasks, particularly in complex spatial regions with occlusions and ambiguous boundaries.

- 3) To verify Contribution 3, we investigated the effect of the SPD module on the student decoder. We performed a controlled comparison between SPD and three variants: a) student-only training without distillation; b) feature-level mimicry using L2 loss; and c) attention-transfer-based guidance. Additionally, we compared SPD with and without self-guided prompting (SPD W/O SH) to evaluate the importance of dual-source prompting.

As shown in Table VI and Fig. 8, SPD significantly enhanced the semantic abstraction capability of the student network, particularly in small-object segmentation and class-consistent region prediction. Its dual-source

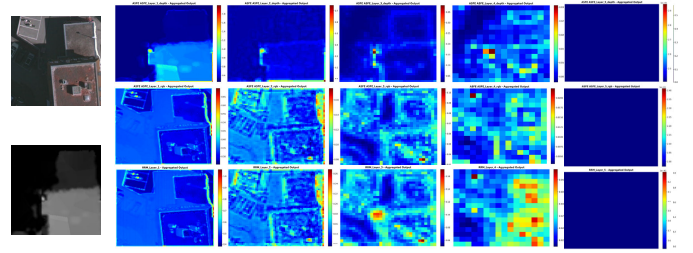


Fig. 9. Module effectiveness visualization.

TABLE VII

QUANTITATIVE EVALUATION OF MODELS ON THE RGB-D MIRROR DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND GREEN, RESPECTIVELY

Method	IOU↑	BER↓	MAE↓
CAINet	74.93	10.42	0.047
MirrorNet	68.37	8.66	0.062
PDNet	77.77	7.77	0.042
SANet	78.43	8.16	0.041
SATNet	78.42	9.31	0.031
SPGSNet-S*	77.49	8.59	0.039

prompting scheme was more effective than conventional attention transfer or feature mimicry strategies. These results confirm that SPD not only improves segmentation performance for small-scale objects but also enhances class-wise semantic consistency, directly supporting Contribution 3.

- 4) *Quantitative Analysis of Module Contributions:* Fig. 9 visualizes the intermediate outputs of our proposed ASFE module and RRM across different network layers. The first column shows the input RGB and nDSM images, respectively. The top two rows display attention maps generated by the ASFE modules, demonstrating adaptive spatial feature enhancement with layer-specific focus patterns. The bottom row shows fused features from the RRM, illustrating robust cross-modal representation learning. These visualizations clearly demonstrate the effectiveness of our proposed modules in enhancing feature discrimination and fusion.

Results of these ablation experiments indicate that each module and distillation strategy we designed successfully addresses the corresponding challenges and that they complement each other. This synergy enables our model to achieve outstanding performance and handle various complex scenarios.

E. Performance on Other Datasets

To further validate the robustness and generalization capability of our proposed student model (SPGSNet), we conducted comprehensive experiments on the RGB-D mirror dataset comprising 3049 RGB-depth image pairs—2000 for training and 1049 for testing. For performance evaluation, we used three key metrics—IOU, mean absolute error (MAE), and balanced error rate (BER)—to assess segmentation accuracy. As summarized in Table VII, SPGSNet-S* achieved an IoU of 77.49, a BER of 8.59, and an MAE of 0.039 on the RGB-D mirror dataset, demonstrating competitive performance across

all metrics. Although it did not rank first individually, our method exhibited consistently stable results with small performance gaps (0.94 in IoU, 0.82 in BER, and 0.008 in MAE) with respect to the corresponding best method. This is particularly noteworthy considering that SPGSNet-S* was designed and optimized for remote sensing imagery without any domain-specific adaptation for indoor mirror segmentation. These cross-domain results validate the robustness and generalization capability of our multimodal fusion and distillation framework, demonstrating its potential applicability beyond the primary target domain of aerial and satellite imagery.

F. Main Research Findings

Experimental results demonstrate that SPGSNet-S* achieved a performance comparable to those of previous approaches while significantly reducing computational complexity, with an mIoU of 80.92% on Vaihingen and 75.32% on Potsdam using only 8.89 M parameters (an 86% reduction in number of parameters compared to that of the teacher model) and 2.29G FLOPs, with minimal performance degradation (1% mIoU drop).

G. Strengths and Comparative Advantages

Compared to existing methods, our approach offers three distinctive advantages. First, ASFE + RRM addresses nDSM noise and heterogeneity more effectively than simple concatenation or attention-based fusion [17], [23], [24], [25] through explicit bilateral filtering, spatial attention, and multiscale recalibration. Ablation studies confirmed that ASFE + RRM improves mIoU by 5.56% over the baseline. Second, GCSD performs graph reasoning directly on 2-D feature maps, avoiding the spatial topology loss from vectorization in prior graph-based distillation methods [34], [35]. This preserves critical boundary information for complex remote sensing structures. Third, SPD dynamically generates visual semantic prompts without textual supervision, addressing the fundamental challenge that text annotations are unavailable in remote sensing—a limitation in existing PL methods relying on vision–language models [19], [20], [21], [37], [38], [39].

H. Limitations and Future Directions

Despite significant progress, several limitations warrant future investigation. First, validation was based only on urban datasets; generalization to agricultural or forestry scenarios requires further verification. Second, the framework is designed for CNN architectures (ResNet/MobileNet); adaptation to Vision Transformers [9], [10], [11] or Mamba-based models needs exploration. Third, edge device deployment and real-time performance have not been thoroughly evaluated. Fourth, the two-stage training paradigm (teacher pretraining then distillation) could be simplified through end-to-end joint training or self-distillation.

Future research directions include: 1) extending the framework to multimodal data beyond RGB–nDSM (hyperspectral, SAR); 2) developing adaptive distillation strategies that adjust transfer intensity based on task difficulty; 3) investigating

neural architectures for optimal lightweight designs; and 4) exploring few-shot learning for rapid adaptation with limited labels.

V. CONCLUSION

This study proposed SPGSNet-S*, a lightweight semantic segmentation framework that addresses three critical challenges in remote sensing—modality heterogeneity in RGB–nDSM fusion, structural information loss in KD, and semantic guidance without textual supervision. Our main contributions include: 1) a dual-branch encoder (ASFE+RRM) that handles nDSM noise through bilateral filtering and spatial attention while enhancing RGB features via graph convolutional context modeling; 2) a GCSD module that preserves spatial topology by operating directly on 2-D feature maps; and 3) an SPD module that dynamically generates visual prompts from teacher features without text annotations. Experimental validation demonstrated that SPGSNet-S* achieved an 80.92% mIoU on the Vaihingen dataset with only 8.89 M parameters and 2.29G FLOPs, representing 86% and 89% reductions in parameter number and computational burden, respectively, compared to those of the teacher while maintaining only 1% performance degradation.

In the future, we will focus on extending our framework to additional modalities, adapting to emerging architectures, and enabling few-shot learning for rapid deployment in new scenarios.

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